

Current-Source Inverter Integrated Motor Drives Using Dual-Gate Four-Quadrant Wide-Bandgap Power Switches

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Abstract—Four-quadrant (FQ) switches that have both bidirectional current and reverse-voltage blocking (RVB) capabilities are appealing candidates for use in current-source inverters (CSI) to significantly reduce the switch conduction losses. However, the FQ characteristics of these switches can create problems during switching events by causing transient overcurrent pulses that damage the inverter. Commutation methods for FQ switches in CSI topologies that address these challenges are presented in this article. Several key aspects of the performance and implementation of these commutation methods are explored for a high-power density 3 kW integrated motor drive incorporating both a current-source active rectifier and inverter stage that has been developed using these FQ switch methods. Experimental results demonstrate the benefits of implementing these commutation methods in CSIs that use FQ switches. The benefits include substantial efficiency improvement compared to the conventional CSI implemented with series power diodes as well as reduced mass and volume of passive components due to the high switching frequency (100 kHz).

Index Terms—Current-source inverter (CSI), four-quadrant (FQ) switches, integrated motor drive (IMD), monolithic switches, wide-bandgap (WBG) devices.

I. INTRODUCTION

PERFORMANCE capabilities of conventional silicon (Si) power devices are reaching their limits due to theoretical limits imposed by the Si material. In contrast, the emergence of wide-bandgap (WBG) power devices is appealing for the future of power electronics by allowing high-efficiency operation at

switching frequencies that can be an order of magnitude higher than allowable values for conventional Si devices, and at higher junction temperatures [1]. Such characteristics provide the basis for motor drive size reduction by shrinking passive components and the cooling system. The improvements made possible by WBG switches are extremely valuable for integrated motor drives (IMDs) that can play a key role in the replacement of fixed-speed drives with adjustable-speed drives in many electric motor applications leading to increased system efficiency [2].

The emergence of power devices with new semiconductor properties encourages a re-evaluation of the power converter configurations that are best suited to use them. Voltage-source inverters (VSIs) have been traditionally preferred overcurrent-source inverters (CSIs) because the widely-used IGBTs and MOSFETs do not inherently provide reverse-voltage blocking (RVB) capability, which is required by the CSI topology. However, recent studies have shown that the VSI topology imposes some important limitations when Si-based switches are replaced by new WBG-based switches, such as motor terminal overvoltages and increased electromagnetic interference (EMI) [3]. In VSIs, the interturn overvoltage can reach values as high as two times the dc-link voltage [4]. In contrast, electric motors excited by CSIs are not subjected to terminal overvoltages due to the presence of the CSI's output filter capacitors. Therefore, machines designed for current source-fed systems can use thinner interturn insulation, improving the slot fill factor and, consequently, the machine's power density [5].

In [6], a power density comparison of different drive topologies for a navy shipboard system was conducted. Stringent conducted EMI performance requirements set by the CE102 standard in the range of 10 kHz to 10 MHz were considered. The study demonstrated that a variable-speed drive comprising a VSI and a voltage-source rectifier (VSR) requires a volume approximately 45% higher than a system comprising a CSI and a current-source rectifier (CSR). The major contributor for this volume discrepancy was the requirement for a much larger common-mode filter in the VSI-VSR topology.

Other advantages of CSIs include their boost capability. For example, the authors of [4] have compared the performance of electric machines specifically designed to be driven by VSIs and CSIs. The machines were targeted for high-speed aircraft compressor drives. It was shown that the machine designed for CSIs was able to achieve higher efficiency (97.7%) and

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almost twice the power density (5.9 kW/kg) of the best machine designed for voltage source-based drives (97.3%, 3.1 kW/kg).

Therefore, the long-neglected CSI topology offers advantages by reducing EMI noise, avoiding motor terminal overvoltages, replacing the VSI's temperature-limited dc-link capacitor with a more robust dc-link inductor, providing boost capability, and improving voltage and current total harmonic distortion at the machine terminals [2], [3].

One of the major drawbacks of CSIs is the requirement to connect a power diode in series with the switching device (switch + diode) to provide RVB capability, which significantly increases the inverter's conduction losses. Devices with natural RVB capability can substantially improve the CSI's efficiency by eliminating the series diodes. When WBG devices are designed as lateral devices (typical for gallium nitride (GaN) devices, but also conceivable in SiC [7]), their channels are symmetrical, making them candidates for monolithic devices with four-quadrant (FQ) capabilities that include RVB properties.

In [8], an appealing monolithic bidirectional switch is designed and fabricated in a common-drain configuration, which substantially reduces its on-resistance and size. In [9], a different monolithic FQ switch topology that also adopts a common-drain structure is proposed. The conduction loss in a CSI power stage that uses monolithic FQ switches in common-drain topologies that share their drift regions can achieve approximately half the conduction loss of a CSI using hybrid switches [10] and even a lower fraction when compared to the baseline CSI using the switch + diode configuration. The use of FQ switches in CSIs also makes it possible to reverse the dc-link current polarity. This feature is particularly advantageous in systems with distributed dc voltage architectures, where the dc-link current polarity can be used to reverse the power flow direction, eliminating the need for full-bridge front-ends [10]. However, FQ switches present challenges during switching events due to problems caused by either dead-time or conduction overlap when commutating the load current [11]. Therefore, the introduction of FQ switches into CSIs requires further investigation since it cannot serve as a simple drop-in replacement for the baseline switch + diode configuration.

FQ switches have received more attention during recent years due to their appeal for use in several power converter topologies, including the matrix converter, Vienna rectifier, T-type multi-level converters, and cyclo-converters. For example, Panasonic has already released research samples of GaN-based monolithic FQ switches that can block voltage >1300 V in both polarities while providing ON-state resistance values similar to those of conventional switches [12].

Recent studies have evaluated many implementation issues associated with WBG switches, including gate drive circuits, dead-time, and parasitic effects [13]–[15]. However, only a few papers have evaluated FQ switches implemented with WBG devices and their use in CSIs. In [3], potential benefits of using FQ switches in CSIs, such as improved efficiency, are identified. Although [3] includes the characterization of a hybrid FQ switch using discrete devices, the article does not focus attention on the issues associated with using FQ bidirectional switches, and no full inverter operation results are provided. In [10], monolithic

FQ switches are used to improve a buck-boost current-source inverter's efficiency, power density and allow bidirectional power flow; however, the article is mainly focused on analyzing a new modulation strategy that reduces the conduction losses and switching losses by shaping the dc-link current. In [16], a new fault-tolerant CSI topology using FQ switches is proposed. The topology takes advantage of monolithic FQ switches to achieve redundancy, fault detection, and loss reduction via current sharing between parallel devices. In [17], FQ switches are employed in the CSR stage of an indirect matrix converter; however, the article is mainly focused on control strategies for operation under unbalanced grid voltage conditions. Neither [10], [16], or [17] analyze the commutation of FQ switches in CSIs and strategies to minimize their losses.

The major objective of this article is to demonstrate that the CSI topology in combination with FQ switches can provide high efficiency and high-power density, making this topology a serious candidate for adoption in IMDs. This article presents, analyzes, and implements commutation methods tailored for applying FQ switches in CSIs that achieve safe and high-efficiency operation. The analytical results are experimentally verified using a 3 kW CSI-IMD demonstrator unit exciting a permanent magnet machine.

The article is structured as follows. In Section II, an explanation of the advantages of using FQ switches in CSIs is provided and different FQ switches realizations are discussed. Section III provides a detailed explanation of the four-step commutation method for CSIs using FQ switches. Key implementation issues are also discussed in detail. Section IV introduces a three-step improved commutation method that reduces the switching loss by taking advantage of the knowledge of the output capacitors' voltage polarity. Section V presents extensive experimental results, including results obtained from a bench-top CSI driving an *RL*-load, and results from a CSI-IMD exciting a surface permanent magnet (SPM) motor. Finally, Section VI uses the experimental results to create efficiency maps for alternative topologies and to estimate the efficiency of future CSI-IMDs that employ monolithic bidirectional switches.

II. HYBRID SWITCH REALIZATIONS

As discussed in the preceding section, the use of monolithic FQ switches is notably beneficial for CSIs since it can significantly reduce conduction losses. However, such monolithic devices are still commercially immature. In their absence, hybrid switches consisting of two discrete devices connected in anti-series can be used to mimic the behavior of future monolithic FQ devices. Four possibilities for the hybrid switch realization using anti-series WBG switches are presented in Fig. 1. Fig. 1(a) and (b) use WBG devices such as lateral GaN high-electron-mobility transistors (HEMTs) that do not incorporate an intrinsic body diode. Two of these devices can be connected in either a common-drain [Fig. 1(a)] or common-source configuration [Fig. 1(b)], both with dual gate drives. Alternatively, WBG devices such as vertical SiC MOSFETs that include body diodes can be used in either the common-drain or common-source configurations, as shown in Fig. 1(c) and (d), respectively. Note

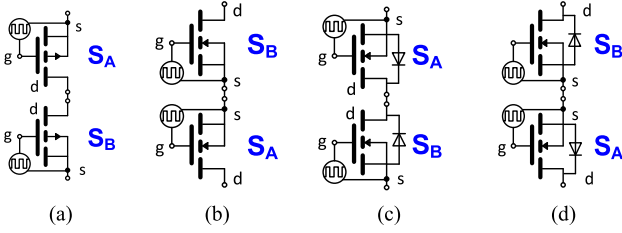


Fig. 1. FQ hybrid switches using two anti-series devices without body diodes. (a) Common-drain configuration. (b) Common-source configuration. FQ hybrid switches using anti-series devices with body diodes. (c) Common-drain configuration. (d) Common-source configuration. [18].

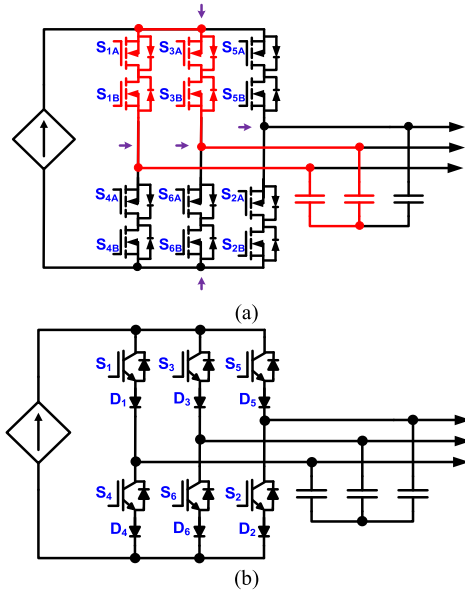


Fig. 2. (a) Three-phase H6-CSI topology implemented with common-source FQ switches [Fig. 1(c)]. (b) Three-phase H6-CSI topology with a diode in series with each switch (IGBT + diode) [18].

that the labels S_A and S_B are reversed for the common-source configurations compared to the common-drain configurations to make all of the analyses in this article generic for both hybrid switch configurations.

Fig. 2(a) shows the baseline H6-CSI topology implemented with six hybrid FQ switches that use the WBG devices with body diodes in the Fig. 1(c) common-drain configuration. The main advantage of the common-drain configuration is that the drift region can be shared by both switches, reducing the total on-resistance of a monolithic FQ switch. However, this advantage does not apply to hybrid switches using two discrete devices.

Another advantage of the common-drain configuration is the use of only five isolated power supplies per inverter to energize the gate drives of all 12 CSI switches. The locations of the common nodes for these five power supplies are identified by purple arrows in Fig. 2(a). Different switches can share a single isolated power supply provided that their source terminals are connected to the same node. In comparison, the common-source configuration requires six isolated power supplies per inverter. In this case, each hybrid switch requires its own isolated power supply connected to the common-source terminal of the hybrid

switch. Extra care must be taken when using shared isolated power supplies to avoid the introduction of parasitic inductances between each switch's source and gate terminals.

The channel in any MOSFET is also capable of conducting current in the reverse polarity whenever its gate voltage is above its threshold voltage. This operating mode, known as synchronous rectification, is commonly used with standard Si MOSFETs to reduce the switch losses during reverse current conduction by shifting the current from the lossy diode to the low-resistance channel. Even though WBG devices such as lateral GaN HEMTs do not incorporate an intrinsic body diode, they still behave as diodes, albeit lossy ones that allow the device to conduct reverse current even when the device is gated "OFF" (e.g., $V_{gs} = 0$) [1], [19]. When the gate voltage is raised above its positive threshold voltage, the reverse voltage across the device falls to the ON-state channel resistive voltage drop corresponding to synchronous rectifier operation. Therefore, a lateral GaN HEMT or FET that does not include body diodes can still be used in hybrid FQ switch topologies, as shown in Fig. 1(a) and (b). Since a conducting channel is playing the role of the reverse-conducting diode when the device is gated "ON," the switch does not exhibit the reverse recovery characteristics associated with body diodes because no minority carriers are involved.

III. COMMUTATION OF CSI IMPLEMENTED WITH FOUR-QUADRANT SWITCHES (FOUR-STEP COMMUTATION)

Fig. 2(b) shows the traditional three-phase CSI topology using RVB switches that are realized by adding a diode in series with IGBTs. The following constraints must always be satisfied in a CSI. (a) There must always be a current conduction path for the dc-link current source. (b) There must be no sudden short circuits applied between any of the inverter output terminals to prevent shorting of the output filter capacitors [20].

Constraint (a) requires that at least one of the switches connected to each of the two dc bus terminals in a CSI must be turned ON at every time instant. To satisfy this first constraint, it is common practice to use an overlap time between an off-going and an oncoming switch connected to the same bus terminal, representing the dual of dead-time in VSIs. On the other hand, condition (b) is always satisfied when using switches in series with diodes, as in Fig. 2(b), because the series diodes will always prevent short-circuit current from flowing between output terminals even if all the switches are turned ON. This is the dual of the role played by anti-parallel diodes in the VSI switches that always provide a path for the inductive load current even if all the switches are turned OFF.

Consider Fig. 2(a) in which the CSI is implemented with six hybrid FQ switches that use the common-drain configuration in Fig. 1(c). It is assumed here that each switch is a WBG MOSFET that includes a body diode, as indicated in Fig. 2(a). In this configuration, meeting constraints (a) and (b) becomes problematic if both switches comprising the hybrid switch (A and B) receive the same gate signals appropriate for a conventional CSI. For example, when commutating from Switch 1 to Switch 3, the output phase capacitors are short-circuited if any overlap is used [shown in red in Fig. 2(a)]. However, this circuit cannot tolerate a

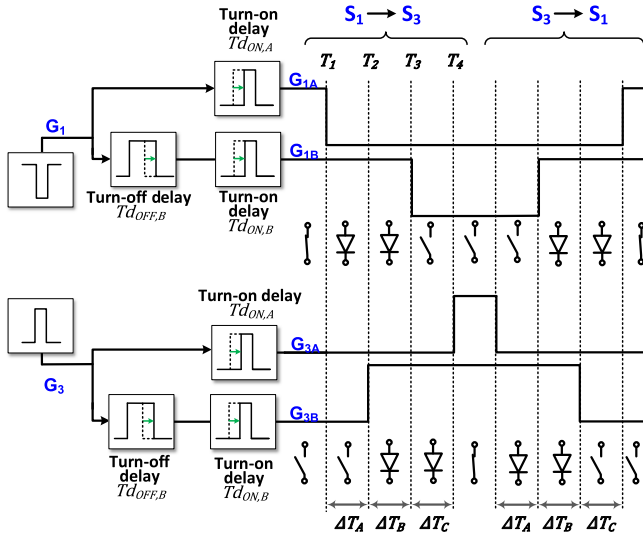


Fig. 3. Four-step commutation for CSIs implemented with FQ switches [18].

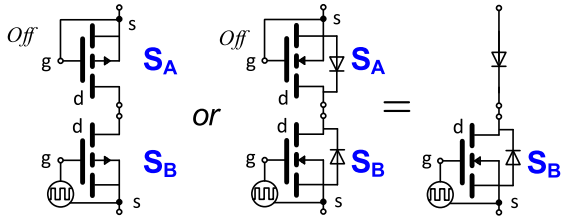


Fig. 4. Equivalency of FQ switch (left) and RVB switch (right) when v_{gs} of SA equals zero.

dead-time either during this commutation which would interrupt the current path for the current-source. Ideally, this problem would be prevented by turning ON one hybrid switch at exactly the same time that the other one turns OFF. However, switch transitions never occur instantaneously due to a number of practical factors, including propagation delays, finite transition times, and parasitic elements, making this ideal synchronism impossible in practice.

The commutation problem can be overcome by adopting the four-step commutation method shown in Fig. 3, which relies on a synchronized switching concept. This method can be understood by first noting that when switch S_A is turned-OFF, this FQ switch behaves as the RVB switch implemented with a diode in series with the switch, as shown in Fig. 4. Hereafter, this FQ switch state will be referred to as switch-diode mode. Therefore, safe operation of the CSI can be guaranteed by turning OFF switch S_{1A} at the beginning of the commutation sequence (time instant T_1 in Fig. 3) and then generating an overlap interval (from T_2 to T_3) during, which switches S_{1B} and S_{3B} are both gated ON. Finally, switch S_{3A} is turned ON at T_4 to reduce the conduction losses of Switch 3 by transferring the switch S_{3A} current from its body diode to its low-resistance channel. The time interval between the turn-OFF of switch S_{1A} and the turn-ON of switch S_{3A} (from time T_1 to T_4) can be interpreted as a dead-time since it prevents the output capacitors from experiencing a transient short-circuit condition.

For completeness, Fig. 3 also shows the transition sequence when Switch 3 is turned OFF and Switch 1 is turned ON again, showing on the right side of Fig. 3 how the switching events are symmetrically reversed in time sequence from the preceding discussion. This commutation method can be easily implemented with turn-ON and turn-OFF delays, as shown in the left side of Fig. 3 where the signals G_1 and G_3 are the gate signals that assume ideal switches (without the requirement for overlap or dead time). Note that the required gate delay logic is the same for every hybrid switch, simplifying the implementation of the commutation method. The step-by-step commutation presented in Fig. 3 is also shown in a sequenced circuit diagram in Fig. 5 to facilitate understanding. In Fig. 5, it is assumed $V_{ab} > 0$.

To ensure the safe operation of the CSI, the logic expressions (1)–(3) must always be true ($= 1$). These logic expressions are only being shown for Switches 1, 3, and 5, but the same conditions must always hold true for Switches 2, 4, and 6. The first expression (1) ensures that a current path for the current source is always provided. If (1) is true, the second expression (2) ensures that no short-circuit of the output capacitors will occur by not allowing more than one A switch to be turned-ON at the same time. Finally, the last expression ensures correct synchronism by not allowing an A switch to be ON if the corresponding B switch is not already ON.

$$(G_{1B} + G_{3B} + G_{5B}) \quad (1)$$

$$(\overline{G_{1A} \cdot G_{3A}}) + (\overline{G_{1A} \cdot G_{5A}}) + (\overline{G_{3A} \cdot G_{5A}}) \quad (2)$$

$$(\overline{G_{1A} \cdot \overline{G_{1B}}}) + (\overline{G_{3A} \cdot \overline{G_{3B}}}) + (\overline{G_{5A} \cdot \overline{G_{5B}}}) \quad (3)$$

These expressions can be used to design a fast-response sequence logic circuit that is placed immediately in front of the gate drives for detecting and preventing fault conditions. It can also be used to help check the gate signals during the design stage by quickly detecting any gate signal generation fault. Alternatively, the data from an oscilloscope can be recorded and (1)–(3) can be evaluated in a data processing tool such as MATLAB to verify the correctness of the gate signals.

Expressions (1)–(3) do not consider the turn-ON (T_{on}) or turn-OFF (T_{off}) times of the switches. Assuming that all the switches of the inverter have the same turn-ON and turn-OFF time, the following relations (4) and (5) must also be obeyed:

$$\Delta T_A, \Delta T_C > T_{off} \quad (4)$$

$$\Delta T_B > T_{on}. \quad (5)$$

This is equivalent to (6)–(8) that are expressed in terms of the three delay times that appear in Fig. 3:

$$Td_{ON,B} > T_{off} \quad (6)$$

$$Td_{OFF,B} > Td_{ON,B} + T_{on} \quad (7)$$

$$Td_{ON,A} > Td_{OFF,B} + T_{off}. \quad (8)$$

If the time spent in the OFF state by the original gate signal is smaller than $T_{d\ OFF,B}$, the B switch will not even be turned OFF. This may somewhat distort the output waveforms, but it does not pose any risk for safe operation since the current-source path is still provided. Similarly, if the ON state command is smaller than

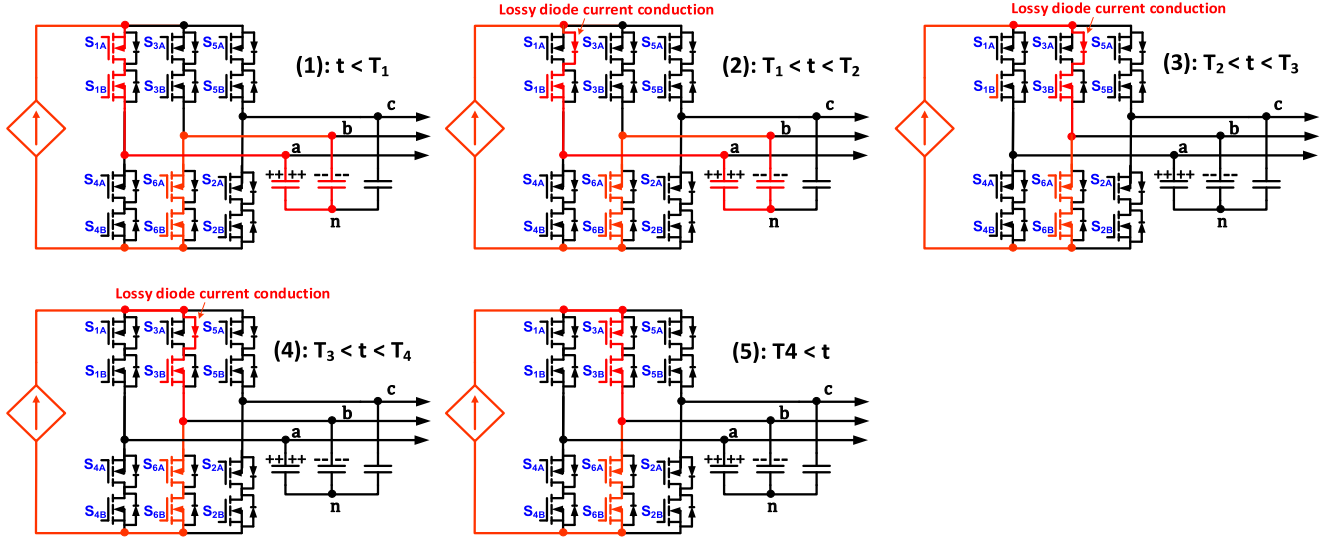


Fig. 5. Step-by-step commutation method for CSIs implemented with FQ switches using four-step algorithm, showing commutation from S_1 to S_3 conduction.

$T_{d\text{ ON},A}$, the A switch will not even be turned ON, which again does not represent a risk, but only contributes to a slight increase in the conduction losses since no synchronized conduction is performed. In order to minimize these distortions, duty cycle compensation can be introduced in a similar manner to how this compensation is used in VSIs.

IV. IMPROVED COMMUTATION OF CSI IMPLEMENTED WITH FOUR-QUADRANT SWITCHES BASED ON OUTPUT VOLTAGE MEASUREMENT (THREE-STEP COMMUTATION)

The approach taken in the preceding section has the advantage of guaranteeing the safe commutation of the FQ switches independently of the output capacitor voltages. However, to ensure such independence, both FQ switches involved in a commutation must operate in the switch-diode mode during two intervals within one complete commutation, as previously shown in Figs. 3 and 5.

In this section, an improved commutation method for FQ switches in CSIs is introduced. This method takes advantage of knowledge of the output capacitor voltages to reduce the number of required intervals that the FQ switches must operate in the switch-diode mode, improving the inverter efficiency. Note that even though the switch-diode mode occurs for short periods, it has conduction losses almost five times higher than the losses when the channel is conducting.

Assume that the CSI output capacitor voltages are polarized as shown in Fig. 7, i.e., $V_{ab} > 0$. In this case, the commutation from Switch 1 to Switch 3 can be safely realized while respecting constraints (a) and (b) presented in Section III, as shown in Figs. 6 and 7, where only the FQ switch S_1 is required to operate in the switch-diode mode. Note from Fig. 7 that, although switch S_1 is in the switch-diode mode during both states (2) and (3) (see Fig. 7), it only conducts current during state (2), since during state (3) the S_{1A} body diode is reverse-biased. Therefore, the three-step commutation method has only one interval per commutation where the lossy body diode operates.

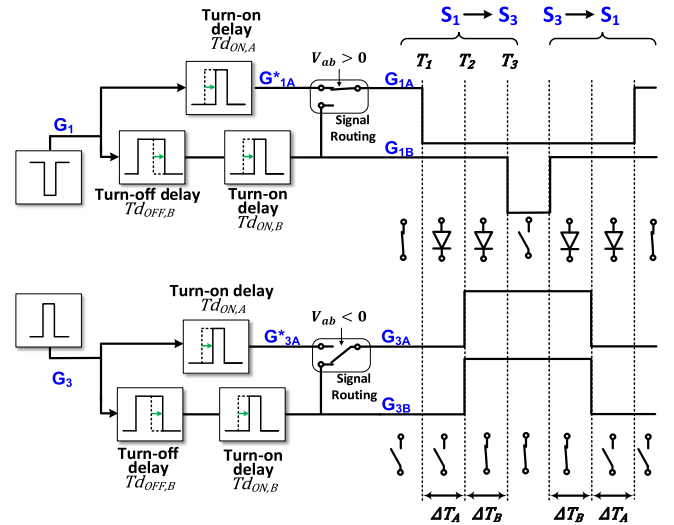


Fig. 6. Three-step commutation method for CSIs implemented with FQ switches.

This represents a 66.7% reduction of the switch-diode mode operating time compared to the previous method where the lossy body-diode operates during three intervals per commutation, as shown previously in Fig. 5.

Equation (9) summarizes the rules that generalize the three-step commutation to all possible states of output capacitor voltages and all combinations of involved switches. In (9), x and y are the phases associated in a commutation event and the gate signals are related to the inverter switches according to Table I. The superscript $*$ refers to the gate signal before the routing switch, as shown in Fig. 6.

The rules in (9) can be explained as follows. When the commutation is among the top switches (numbered as 1, 3, and 5), the hybrid switch connected to the output phase with the lower voltage can have a single signal for both gates. When the commutation is among the bottom switches (numbered as 2, 4,

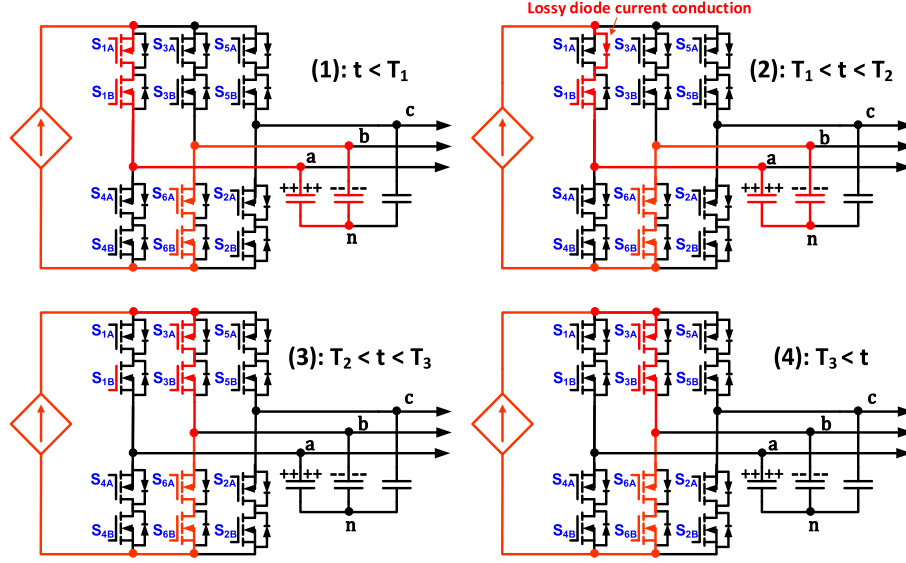


Fig. 7. Step-by-step commutation method for CSIs implemented with FQ switches using three-step algorithm, showing commutation from S_1 to S_3 conduction for $V_{ab} > 0$.

TABLE I
CORRESPONDENCE OF GATE SIGNALS AND SWITCHES

Gate Signal	Discrete Switch	Gate Signal	Discrete Switch
$G_{a,top,swA}$	G_{1A}	$G_{a,top,swB}$	G_{1B}
$G_{b,top,swA}$	G_{3A}	$G_{b,top,swB}$	G_{3B}
$G_{c,top,swA}$	G_{5A}	$G_{c,top,swB}$	G_{5B}
$G_{c,bot,swA}$	G_{2A}	$G_{c,bot,swB}$	G_{2B}
$G_{a,bot,swA}$	G_{4A}	$G_{a,bot,swB}$	G_{4B}
$G_{b,bot,swA}$	G_{6A}	$G_{b,bot,swB}$	G_{6B}

and 6), the hybrid switch connected to the output phase with the higher voltage can have a single signal for both gates. For the sake of example, Fig. 6 considers that $V_{ab} > 0$; therefore, in accordance to (9), $G_{a,top,swA} = G_{a,top,swB} = G_{1A} = G_{1A}^*$ and $G_{b,top,swA} = G_{b,top,swB} = G_{3A} = G_{3B}^*$.

$$\begin{aligned} \text{Case } (V_{xy} > 0) & \begin{cases} G_{x,top,swA} = G_{x,top,swA}^* \\ G_{y,bot,swA} = G_{y,bot,swA}^* \\ G_{y,top,swA} = G_{y,top,swB} \\ G_{x,bot,swA} = G_{x,bot,swB} \end{cases} \\ \text{Case } (V_{xy} < 0) & \begin{cases} G_{x,top,swA} = G_{x,top,swB} \\ G_{y,bot,swA} = G_{y,bot,swB} \\ G_{y,top,swA} = G_{y,top,swA}^* \\ G_{x,bot,swA} = G_{x,bot,swA}^* \end{cases} \end{aligned} \quad (9)$$

The left side of Fig. 6 also shows a simplified schematic illustrating how to implement this commutation strategy using turn-ON and turn-OFF delays and a signal routing switch. The schematic is remarkably similar to the one in Fig. 3, except for the introduction of the two signal routing switches for the gate signals that drive the A-switches. The signal routing switch operates according to the conditions specified by (9). The status of these switches can be updated every PWM cycle and (1)–(5) still needs to hold true in the three-step commutation.

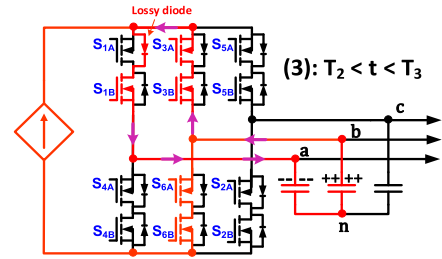


Fig. 8. Illustration of fault condition during commutation from S_1 to S_3 conduction using the three-step algorithm when the voltage sensor measures $V_{ab} > 0$ in error because of an offset error in the sensor.

Voltage sensors offset error can lead to a wrong measurement of the polarity of the output capacitor's voltage, particularly when the voltage waveform is close to zero crossings. Suppose that the output capacitors are charged in the opposite direction, as shown in Fig. 8, but the commanded gate signals remain unchanged as presented in Fig. 6 because of an offset error in the voltage measurements that causes the sensor to mistakenly report that $V_{ab} > 0$. In this case, the output capacitors of phase a and b would be shorted, as shown by the purple arrows in Fig. 8. Therefore, the three-step commutation method can fail if the output capacitor's voltage measurement has an offset error. To avoid the fault risk posed by this voltage offset condition, the three-step commutation can be combined with the four-step commutation method presented in Section III. More specifically, the four-step commutation method can be activated whenever the measured voltage magnitude is within the sensor's maximum offset voltage error since the four-step algorithm is insensitive to the line-to-line voltage polarity.

The four-step commutation algorithm always meets constraints (a) and (b) of Section II, independent of the output capacitor voltage state. When the voltage measurement falls inside the sensor error margin, the four-step commutation method is used.

TABLE II
ACTIVE AND ZERO STATES FOR DIFFERENT SVPWM SECTORS

Sector	State A	State B	Zero State Option 1	Zero State Option 2
I	S_1, S_2	S_6, S_1	S_3, S_6	S_5, S_2
II	S_2, S_3	S_1, S_2	S_1, S_4	S_3, S_6
III	S_3, S_4	S_2, S_3	S_5, S_2	S_1, S_4
IV	S_4, S_5	S_3, S_4	S_3, S_6	S_5, S_2
V	S_5, S_6	S_4, S_5	S_1, S_4	S_3, S_6
VI	S_6, S_1	S_5, S_6	S_5, S_2	S_1, S_4

Therefore, using this hybrid method, the conditional inequalities that appear in (9) are modified to be:

Case $V_{xy} > e_{\text{margin}} \rightarrow$ Three - step operation

Case $V_{xy} < -e_{\text{margin}} \rightarrow$ Three - step operation. (10)

Equation (11) defines when four-step commutation is applied:

Case $(-e_{\text{margin}} < V_{xy} < e_{\text{margin}}) \rightarrow$ Four - step op. (11)

A. Operation of the Three-Step Commutation When Using Space Vector Pulse Width Modulation (SVPWM)

When using the SVPWM, only four switches are used per sector. For example, during sector II, only switches 1, 2, 3, and 6 operate if the zero-state is realized by turning ON switches 3 and 6. Therefore, during this sector, the commutation among the top switches is always independent of switch 5, which is continuously OFF. Similarly, the commutation among the bottom switches is also independent of switch 4. Consider Fig. 7, which shows a commutation event that occurs within Sector II of the SVPWM. Given the fact that switch 5 never operates within this sector and given the voltage polarization of the output capacitors $V_{ab} > 0$, the discrete switch S_{3A} can be left in the ON-state during the entire sector II without posing the risk of shorting the output capacitors. In summary, instead of turning ON and OFF switch S_{3A} in synchronism with the switch S_{3B} as previously shown in Fig. 6, the switch S_{3A} can be kept on during the entire sector II as long as $V_{ab} > 0$. On the other hand, in case $V_{ab} < 0$, the switch S_{1A} would be the one kept continuously on during sector II.

Keeping the A switch continuously ON within the sector interval avoids desynchronization between the A and B switches due to propagation delays. Such desynchronization can lead to the lossy switch-diode mode operation. For example, suppose that the gate signal G_{3A} shown in Fig. 6 is slightly delayed with respect to the gate signal G_{3B} . The delay would lead the hybrid switch to operate for a short interval in the lossy switch-diode mode before the A-switch is turned ON. This problem is prevented if the gate signal G_{3A} can be safely kept continuously ON within the entire sector (given $V_{ab} > 0$).

Table II shows the active states and the zero state options that ensure that, for every sector of the SVPWM, there will always be two switches constantly OFF (one among the top switches and another among the bottom ones). Therefore, as previously explained, the A switch of the top hybrid switch connected to the output phase of lower voltage and the A switch

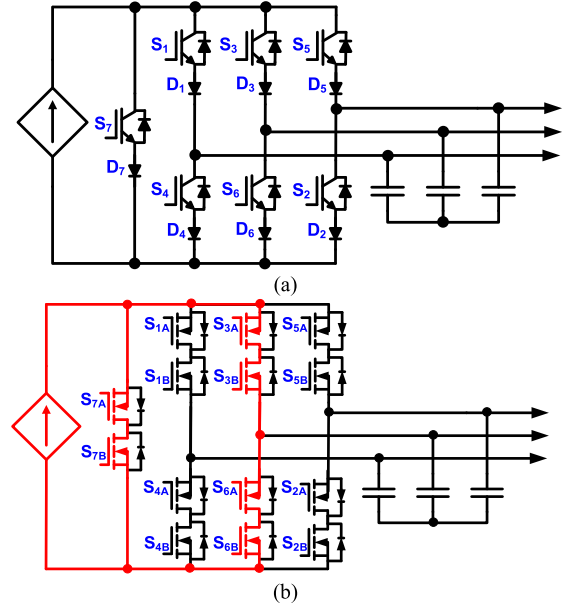


Fig. 9. (a) Three-phase H7-CSI topology using IGBT + diode RVB switches. (b) Three-phase H7-CSI topology implemented with common-drain FQ switches [Fig. 1(c)].

of the bottom hybrid switch connected to the output phase of higher voltage can be kept continuously ON during the entire SVPWM sector interval. Such operation can reduce the CSI losses associated with the switch-diode conduction mode due to desynchronization of the gate signals A and B.

B. Commutation of H7 CSI Implemented With Four-Quadrant Switches

An existing alternative CSI topology that improves the efficiency of the baseline H6-CSI topology is the H7-CSI topology shown in Fig. 9. The additional switch S_7 is used to implement the PWM zero states instead of the S_1 – S_6 switches, allowing these switches to change states under zero-current soft-switching conditions. This section discusses extension of the four-step and three-step commutation methods to this H7-CSI topology when using FQ switches.

The H7-CSI topology using the (switch + diode) configuration for S_1 – S_7 is shown in Fig. 9(a). This topology reduces the conduction losses during zero-states by activating only the S_7 switch instead of both switches in any of the three phase-legs. Fig. 9(b) shows the H7-CSI topology when bidirectional switches are used for the S_1 – S_6 switches. In this topology, the seventh switch must also be implemented as an FQ switch. If a single switch were used without a series diode, the constraint (b) presented in Section III would not be met during active states. More specifically, the output capacitors would be shorted during active states due to conduction of the S_7 body diode.

The S_7 switch can be incorporated into both the four-step and three-step commutation methods, and its operation can be summarized by a fundamental rule for safely operating the seventh switch: if both FQ switches in any inverter phase-leg are completely turned-ON (i.e., a zero state), the S_7 FQ switch

can be active. Applying this rule to the four-step commutation method example in Figs. 3 and 5 means that the S_7 switch will be turned ON at instant T_4 when all four MOSFETs in the S_3/S_6 phase-leg are in their ON-states. Applying this rule to the three-step commutation method example in Figs. 6 and 7 means that the S_7 switch will be turned ON at instant T_2 . As before with the H6-CSI topology, the switching events are symmetrically reversed in time sequence during the turn-OFF process.

Both the S_7 switch and a complete phase-leg of the inverter can be ON at the same time during zero states, as shown in Fig. 9. This reduces the conduction losses by providing two parallel paths for the dc-link current. Assuming that all of the FQ switches have the same channel resistance R , the conduction loss during the zero state can be calculated as

$$P_{\text{cond}} = I_{\text{DC}}^2 (R \parallel 2R) = \frac{1}{3} (2RI_{\text{DC}}^2). \quad (12)$$

That is, the conduction loss during the zero state in the H7 topology is one-third of the H6 zero-state conduction loss. If SVPWM modulation is used, the zero state normalized operating time for one SVPWM sector can be calculated as shown in (13) and (14), according to [21]. Therefore, employing the H7 configuration to reduce the zero state conduction loss up to 66.7% has its largest impact when the modulation index $m_a = 0$, and the impact of this reduction gradually decreases as the modulation index approaches unity, $m_a = 1$.

$$\frac{T_0}{T_s} = 1 - m_a \left(\sin \left(\frac{\pi}{6} - \theta \right) + \sin \left(\frac{\pi}{6} + \theta \right) \right) \quad (13)$$

$$\text{avg} \left(\frac{T_0}{T_s} \right) = 1 - 0.95493 m_a. \quad (14)$$

More information about the H7-CSI topology including an alternative modulation strategy that allows the use of single-gate bidirectional switches can be found in [22].

V. EXPERIMENTAL RESULTS

A. Instrumentation and Efficiency Measurements

Fig. 10(b) illustrates the voltage and current waveforms of a typical CSI when exciting an RL load or a motor. The efficiency is measured from points A to B in Fig. 10(a). This setup measures an unfiltered voltage waveform at the dc-link but has the advantage of excluding the dc-link inductor losses from the measurements.

It has been experimentally verified that for both measurement techniques, less than 0.01% of the input and output power content is associated with frequency components higher than 1 kHz. This conclusion is based on experimental data recorded using a *MDA810A* oscilloscope at a sampling rate of 500 MS/s when using current and voltage probes with bandwidths of 100 MHz (CP031) and 120 MHz (HDV3106), respectively. Since nearly all the power content lies at low frequencies below 1 kHz, power measurement instruments that have high accuracy in this low frequency range are preferred. The *Yokogawa 1806T* power analyzer was selected for the CSI efficiency measurement due to the high accuracy of its voltage and current measurements. According to the equipment manual, the power measurement

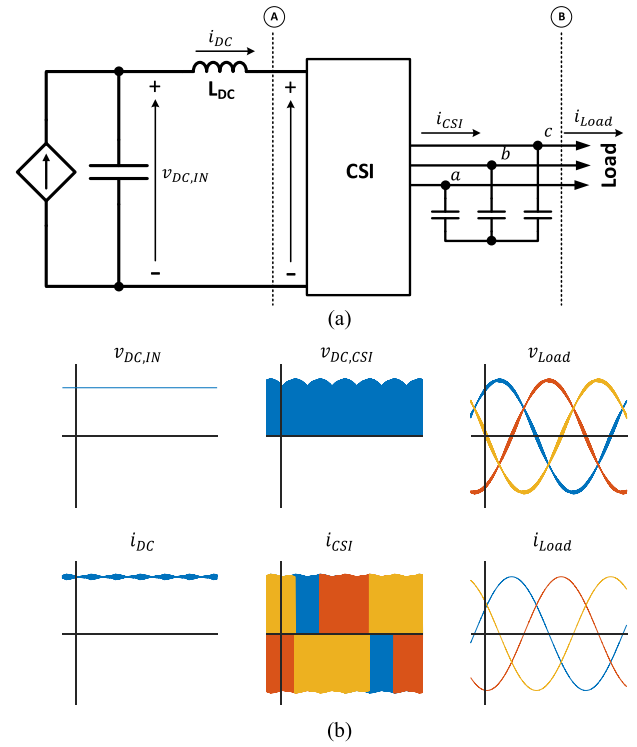


Fig. 10. (a) CSI schematic with measurements identified and power measurement location (A to B). (b) Qualitative voltage and current waveforms of a typical CSI when driving an RL load or a motor.

uncertainty should be less than 0.4% [23]. To improve the precision (but not the accuracy) of the measurements, the samples averaging function was activated while the CSI was operating under steady-state conditions.

It should be noted that the conduction losses associated with the output filter capacitors are small. Based on information provided by the capacitor manufacturer [24], the resistance of each capacitor (*KRM55TR73A224MH01K*) is lower than 70 mΩ in the frequency range from 100 kHz to 10 MHz. This frequency range contains more than 90% of the capacitor current frequency content. In the tested CSI unit, ten capacitors per phase are connected in parallel for each filter capacitor, leading to a total high-frequency resistance of approximately 7 mΩ. Including this resistance value in the simulation leads to a total capacitor for all three phases of less than 0.5 W, which correspond to only 0.015% of the CSI nominal 3 kW power rating.

Finally, only the high-power circuit losses are measured. Therefore, no losses associated with auxiliary circuits such as controller, sensors, and gate drivers were considered either, as those losses can vary substantially depending on the parts used.

B. Commutation of H7 CSI Implemented With Four-Quadrant Switches

To evaluate the performance improvement when using FQ switches, experiments were conducted using the H6-CSI topology implemented with RVB switches (i.e., MOSFET + diode) and the same H6-CSI with hybrid FQ switches. The SiC MOSFETs used to implement the hybrid switches and the RVB switches

TABLE III
CSI BASELINE OPERATION PARAMETERS FOR INVERTER
PERFORMANCE EVALUATION

Parameters	Value
Output phase-phase voltage	196 Vrms
Output power	2 kW
DC-link current	8 A
DC-link inductance	320 μ H
Output filter capacitance per phase (C)	2.2 μ F
Load phase inductance (L_s)	2.95 mH
Load phase resistance (R_s)	19 Ω
Electrical fundamental frequency	100 Hz
Modulation index	1
Gate resistance	10 Ω (on), 2 Ω (off)
Gate voltage V_{GS}	15 V/-3 V
Switching frequency	100 kHz

TABLE IV
PARAMETERS VARIED DURING TESTS FOR INVERTER
PERFORMANCE EVALUATION

Test #	Parameters Varied	Variation Range	Figures
1	Switching frequency	25-100 kHz	Fig. 11(a)
2	DC-link current, switching frequency	0-8 A @ 100 and 25 kHz	Fig. 11(b)-(d)
3	Output phase-phase voltage, switching frequency	0-196 V @ 100 and 25 kHz	Fig. 11(e)-(h)
4	Fundamental frequency	25 and 100 Hz	-
5	Overlap/deadband interval	60-300 ns	Fig. 11(i)
6	Output power (voltage and current)	0-3 kW	Fig. 11(j)-(l)

were all identical (C3M0065090J). The diodes used in the RVB switches were CVFD20065A SiC Schottky diodes. The CSI efficiency and losses were measured at different operating conditions. The objective was to collect data to estimate the behavior of the inverter's switching, conduction, and stray losses with respect to voltage, current, switching frequency, switch configuration, and commutation method. It will be later shown that the stray losses are most likely associated with dielectric loss in the output ceramic capacitors, but this hypothesis requires further investigation that goes beyond the scope of this article.

An effort was made to decouple the impact of the device junction temperature on the results of the tests conducted with both topologies (i.e., MOSFET + diode and hybrid FQ switches). More specifically, the cooling fan's airflow was adjusted until the steady-state case temperature of the hottest MOSFET device measured using a thermal camera during each tested operating condition was approximately the same value (75 °C) measured during baseline operation at 2 kW. The baseline operating conditions are summarized in Table III. Noteworthy, the baseline does not correspond to the nominal power rating of the CSI (3 kW) and an RL load was used to emulate the motor load. At the end of this section, the baseline data will be used to predict the loss distribution under different operating conditions, including nominal rated power operation at 3 kW.

Table IV shows the parameters that were varied for the different tests. The collected experimental data is plotted in

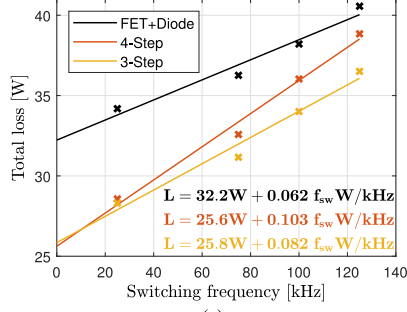
Fig. 11. Three datasets are presented for each test. The first dataset corresponds to the traditional CSI using MOSFET + diode (shown in black), the second and third datasets both use the CSI implemented using hybrid bidirectional common-source switches; however, one dataset corresponds to the four-step operation (shown in red), while the other corresponds to three-step operation (shown in yellow). Note that for Tests #2 and #3, the connected load impedance needs to be adjusted to deliver the desired combinations of output current and voltage levels. For all tests, the CSI is operated at full modulation index ($m_a = 1$) to improve the inverter efficiency. The front-end converter, which acts as a buck stage, regulates the dc-link current magnitude. As shown in Fig. 10, the front-end converter is excluded from the efficiency measurements.

Test #1. Loss for different switching frequencies: The first test verifies the linearity of the inverter losses with respect to the switching frequency for baseline operating conditions. Using the collected data, the inverter loss can be expressed as linear functions of the switching frequency as presented by the in-graph equations in Fig. 11(a), which were obtained via linear regression. From these equations, the conduction loss, which corresponds to the constant term, can be easily segregated from the switching loss. As expected, the FET + diode configuration has the highest conduction loss (32.2 W), while the four-step FET + FET and three-step FET + FET configurations have essentially identical conduction loss (25.6 and 25.8 W, respectively). Theoretically, the four-step FET + FET and the three-step FET + FET conduction loss should be the same since the same power devices are used for both CSIs and the commutation method should only affect the switching loss.

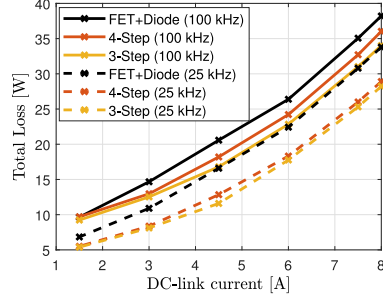
On the other hand, the FET + diode configuration has the lowest switching loss coefficient (0.062 W/kHz), which is again expected since the four-step and three-step methods introduce additional switching loss every commutation event due to the reverse conduction of the body diode during the switch-diode mode as explained in Sections II and III. Finally, the three-step commutation method helps to reduce the switching loss coefficient of the FET + FET configuration from 0.103 W/kHz (four-step) to 0.082 W/kHz (three-step) since the switch-diode mode conduction time during every switching event is reduced.

Since the FET + FET configurations have higher switching losses than the FET + diode configuration, the FET + diode configuration can become the most efficient option if the switching frequency is high enough. For the baseline CSI operating at the conditions provided in Table III, this crossover frequency occurs at 125 and 220 kHz, when using four-step and three-step commutations, respectively. This frequency has been estimated using (19) provided later in the discussion of Test #6. If this crossover switching frequency becomes a barrier to using FQ switches, the availability of faster switching devices should be investigated.

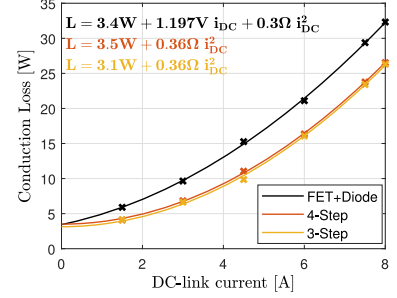
Test #2. Loss sensitivity to current: This test has the objective of identifying the switching and conduction loss dependence on the current level. Therefore, the connected load impedance is varied to modify the dc-link and the output current level while maintaining the same output voltage. The measurement of the

Test #1: Total CSI Loss vs. Switching Frequency @ Baseline Operating Conditions

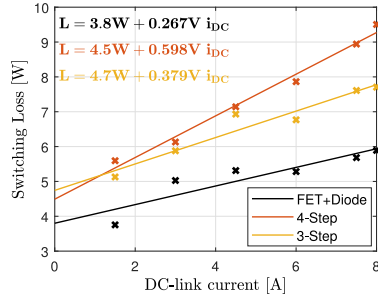
(a)

Test #2: Loss Sensitivity to Output Current @ Output Phase-Phase Voltage = 196 V

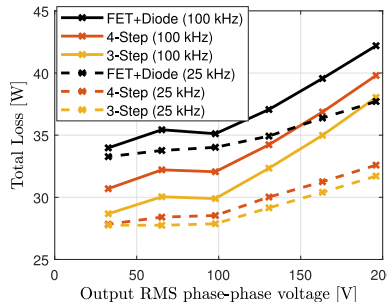
(b)



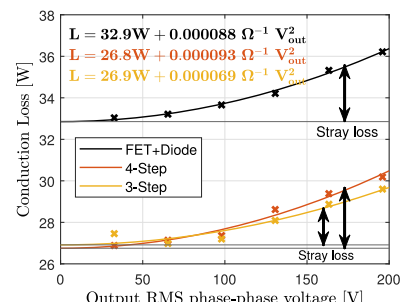
(c)

Test #2 Results (Continued)

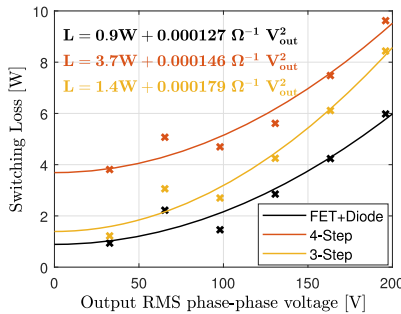
(d)

Test #3: Loss Sensitivity to Output Voltage @ Constant DC-Link Current = 8.5 A

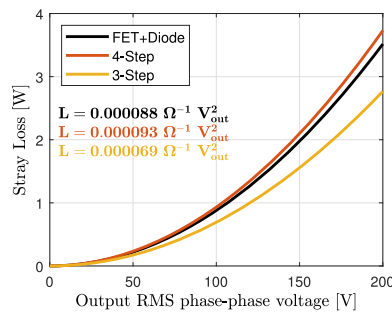
(e)



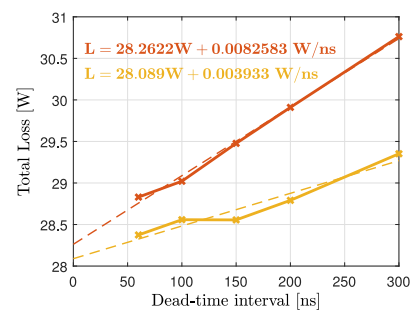
(f)

Test #3 Results (Continued)

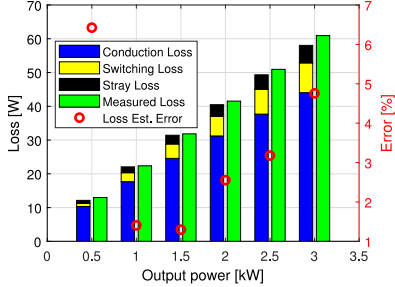
(g)

Test #4: Stray Loss from Test #3 Results

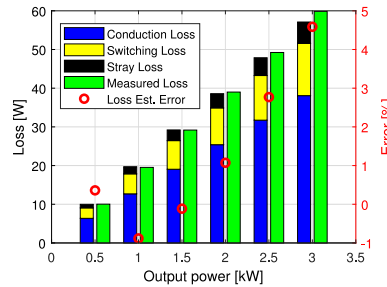
(h)

Test #5: Loss Sensitivity to Dead-Time

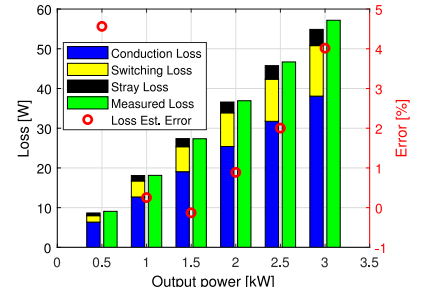
(i)

Test #6: Predicted CSI Efficiency vs. Output Power and Measured Error with Motor Load used for Baseline Operation Conditions (@ 100Hz)

(j)



(k)



(l)

Fig. 11. Experimental results based on operation of the CSI driving an RL-load. (a) Test #1: Losses for different switching frequencies. (b) Test #2: Loss sensitivity to current. (c) Test #2: Conduction loss sensitivity to current. (d) Test #2: Switching loss sensitivity to current. (e) Test #3: Loss sensitivity to voltage. (f) Test #3: Conduction loss sensitivity to voltage. (g) Test #3 Switching loss sensitivity to voltage. (h) Estimated stray loss for different output phase voltages. (i) Test #5: Loss sensitivity to dead-time. (j) Test #6: Output power and losses verification for FET + Diode configuration. (k) Test #6: Output power and losses verification for FET + FET four-step configuration. (l) Test #6: Output power and losses verification for FET + FET three-step configuration.

different losses sensitivity to current is accomplished by first measuring the inverter loss at different current levels for each switching frequency (100 and 25 kHz), as shown in Fig. 11(b). Next, the conduction and switching losses for each current level are decoupled as done in Test #1 to generate the points shown

in Fig. 11(c) and (d). Finally, curve fittings are applied to the obtained points to identify the loss sensitivities to current. As expected, the conduction loss in all three topologies is highly dependent on the current, as shown in Fig. 11(c). Note that, contrary to expectations, the projected conduction loss at zero

current is not zero but approx. 3 to 3.5 W. This suggests that there is another source of loss that generates considerable loss even at low currents. This source of loss should be independent of current and not associated with switching events since, at this extrapolated point, the current and the switching frequency losses both tend to zero. As will be discussed later, the presence of this term is most likely associated with the output capacitor dielectric losses.

Fig. 11(c) shows that in the FET + FET configurations, the conduction loss has a quadratic behavior with respect to the current since the conduction loss is due to the hybrid switch's channel resistances. In the CSI, at any instant, the dc-link current is being conducted by two hybrid switches consisting of four channel resistances. Based on the 0.36Ω resistance coefficient obtained from the curve fitting [see in Fig. 11(c)], the channel resistance of each device can be estimated to be approx. $90 m\Omega$. This resistance value also accounts for other contributors to the conduction losses including the PCB traces, wires, and connectors. In the FET + diode configuration, the conduction loss function has a first-order term associated with the diode's voltage drop and a quadratic term associated with the resistive components contributed by the diodes and switches.

Fig. 11(d) shows that for all configurations, the switching loss sensitivity to the current level is low. Nevertheless, the FET + FET configuration still has a higher dependence on the current level since, for this configuration, every commutation has intervals of switch-diode conduction mode where the losses are dependent on the current level. This dependence is attenuated when using the three-step commutation method, as expected.

Test #3. Loss sensitivity to output voltage: This test bears some similarity to Test #2, but the switching and conduction losses are evaluated as the voltage level is changed. Due to the limitations of the connected load, this test was realized at a fixed current of 8.5 A, slightly above the current level of 8 A used for baseline operation. The results of Test #3 are displayed in Fig. 11(e)–(g). Contrary to expectations, the conduction losses in Fig. 11(f) increase with voltage. Since the conduction loss should not depend on the voltage, this voltage dependence suggests that there is another loss component beyond the switching and conduction losses that has not been incorporated into the model. Based on Fig. 11(f), this loss component increases with voltage, but it is decoupled from switching events, since Fig. 11(f) shows curves that were extrapolated via linear regression where the switching frequency tends to zero. This stray loss will be examined in the next test. Finally, Fig. 11(g) shows that the switching loss increases with voltage in an approximately quadratic manner.

Test #4. Stray losses: As mentioned in the discussion of results for Tests #2 and #3, the inverter appears to have a loss component beyond the switching and conduction losses that is referred to as stray loss. The stray loss for each configuration can be calculated by subtracting the offset of the conduction loss, as shown by the arrows labeled as “stray loss” in Fig. 11(f). The stray losses calculated for each configuration from the Test #3 data is presented separately in Fig. 11(h). Since the inductor is excluded from the loss measurement, this stray loss is most likely associated with the output capacitors. In fact, the capacitor

dielectric losses can be modeled as (15) [25], where K_{diel} is the dielectric loss constant, V_{out} is the CSI output voltage applied across the capacitor and f is the frequency

$$P_{\text{diel}} = K_{\text{diel}} V_{\text{out}}^2 f. \quad (15)$$

To strengthen the argument that the stray losses are associated with the capacitor dielectric loss, another test was conducted operating the CSI FET + diode topology with the same parameters of the baseline operation but at a fundamental frequency of 25 Hz, which corresponds to 1/4 of the original fundamental frequency. Since the voltage, current, and switching frequency are the same, the difference in the losses would correspond to the difference in the capacitor dielectric loss. Noteworthy, at such low fundamental frequencies, the load reactance is negligible and the change in the fundamental frequency has minimal impact on the load impedance and power factor. Based on (15), the dielectric loss at 25 Hz should correspond to 25% of the dielectric loss at 100 Hz. Therefore, since the estimated dielectric loss at 100 Hz is 3.4 W [see Fig. 11(h) @198.4 V], the stray loss should reduce to 0.85 W, which is close to the 1.1 W measured experimentally. Unfortunately, the majority of ceramic capacitor manufacturers do not provide data for evaluating losses associated with low-frequency, large-signal waveforms [25], [26].

The observed stray loss in the inverter and its attribution to the capacitor dielectric loss deserves further investigation, but this topic falls outside the scope of this article. The authors in [25] and [26] provide techniques for measuring large-signal losses in ceramic capacitors. Finally, some of the observed stray losses are likely a consequence of minor unmodeled loss sources. For example, parasitic resistances, inductances, and capacitances can create resonant circuits that are excited nonuniformly at different switching frequencies, generating low-amplitude losses that were not modeled during this investigation.

Test #5. Loss sensitivity to dead-time: This test is meaningful only for the FET + FET topology. The dead-time interval that defines the duration of the switch-diode mode is varied to evaluate the contribution of the switch-diode conduction losses for both the four-step and three-step commutation methods. The results of this test are presented in Fig. 11(i). As previously explained in Section IV, the three-step commutation method enables a reduction of the switch-diode mode loss by 66.7% compared to the four-step method. Therefore, by varying the dead-time interval, it is expected that the loss variation with three-step commutation will be approx. three times lower than the loss variation with the four-step method. However, even though the three-step commutation reduces the time that the dc-link current is being conducted through the lossy switch body diodes to one-third of the time with four-step commutation, this current is redirected to the switch A channel, which also has some ohmic loss.

The expected loss variation with respect to dead-time for the four-step and for the three-step commutation can be expressed as shown in (16) and (17), respectively. In the four-step commutation, the loss variation is completely associated with the interval when the conduction loss is dissipated in the body diode. On the other hand, with three-step commutation, the loss variation

TABLE V
LOSS COEFFICIENTS FOR CSI LOSS ESTIMATION

Coefficient	FET+Diode	FET+FET	FET+FET three-step	Unit
V_{drop}	1.197	0	0	V
$R_{channel}$	0.3	0.36	0.36	Ω
K_{diel}	0.879	0.358	0.266	$\mu\text{Ws}/V^2$
$K_{sw,I}$	0.267	0.598	0.379	Ω
$K_{sw,V}$	127	146	179	$\mu\Omega^{-1}$
$L_{sw@2kW}$	5.79	9.50	8.46	W
$V_{out@2kW}$	198.35	198.35	198.35	V
$I_{out@2kW}$	8.40	8.40	8.40	A
f_{swbase}	100	100	100	kHz

is associated with the body diode conduction loss during a single interval and to the channel conduction loss during two intervals (assuming that the dead-time interval is divided into three identical intervals as previously presented in Section III)

$$\Delta L_{four-step} = K_{cond} \Delta t_{deadtime} I_{DC} V_{BD} \quad (16)$$

$$\Delta L_{three-step} = K_{cond} \frac{\Delta t_{deadtime}}{3} (I_{DC} V_{BD} + 2I_{DC}^2 R_{chan}) \quad (17)$$

Using (16) and (17), the ratio between the slopes of the traces shown in Fig. 11(i) can be predicted to be

$$\frac{\frac{\Delta L_{four-step}}{\Delta t_{deadtime}}}{\frac{\Delta L_{three-step}}{\Delta t_{deadtime}}} = \frac{3V_{BD}}{(V_{BD} + 2I_{DC} R_{chan})} \approx 2.27 \quad (18)$$

where the body diode voltage drop V_{BD} is estimated to be -4 V (based on C3M0065090J datasheet) for a dc-link current I_{DC} of 8 A and the switch channel resistance is estimated to be 80 m Ω (based on Test #2).

As observed in Fig. 11(i), the measured slope with four-step commutation is approximately 2.1 times the slope with three-step commutation. The reason that the experimentally measured ratio between the two slopes is slightly lower than the predicted value can be attributed, at least in part, to the requirement of operating using the four-step commutation in the vicinity of all voltage sensor zero crossings as specified by (10) and (11) in Section IV.

Test #6. Predicted efficiency at different output power and losses verification: Based on the results of the conducted tests, an empirical formula for predicting the CSI losses at different operating conditions can be defined as follows:

$$\begin{aligned} L = & V_{drop} I_{dc-link} + R_{channel} I_{dc-link}^2 + K_{diel} U^2 f \\ & + f_{sw}/f_{sw-base} (L_{sw@2kW} + K_{sw,I} \Delta I_{dc-link} \\ & + K_{sw,v} (\Delta V_{out}^2 + 2\Delta V_{out} V_{out@2kW})). \end{aligned} \quad (19)$$

The coefficients of this formula for each CSI topology were extracted from the curve fittings of Fig. 11 and are summarized in Table V. Equation (19) has three loss components: conduction, switching, and stray loss. The conduction loss is modeled as a combination of diode voltage drop with a channel resistance: $P_{cond} = V_{drop} I_{dc-link} + R_{channel} I_{dc-link}^2$; the stray

loss is modeled as (15) and the switching loss is expressed as shown in (20).

$$\begin{aligned} L_{sw} = & \frac{f_{sw}}{f_{swbase}} (L_{sw,I} (I_{dc-link@2kW} + \Delta I_{dc-link}) \\ & + L_{sw,V} (V_{out@2kW} + \Delta V_{out})) \\ L_{sw} = & \frac{f_{sw}}{f_{swbase}} (L_{sw@2kW} + K_{sw,I} \Delta I_{dc-link} \\ & + K_{sw,v} (\Delta V_{out}^2 + 2\Delta V_{out} V_{out@2kW})) \end{aligned} \quad (20)$$

where

$$\begin{aligned} L_{sw,I} (I) &= K_{sw,I} I \\ L_{sw,V} (V) &= K_{sw,v} V^2. \end{aligned} \quad (21)$$

As shown in (21), the switching loss is modeled as a quadratic and linear dependence on the dc-link current and output voltage, respectively. These expressions were obtained from the sensitivity analysis in Fig. 11(d) and (g) that were developed for 2 kW baseline operation. The conducted sensitivity analysis decouples and approximates the contribution of the output voltage and the dc-link current on the switching losses. The decoupling simplifies the switching loss model and provides accurate results in the vicinity of the baseline operating conditions. The estimation accuracy is degraded as the operating point becomes more distant from baseline operation. Nevertheless, the accuracy of the overall loss model is highly respectable for the entire range of investigated conditions, as discussed in the next paragraph.

Test #6 has two major objectives. The first objective is to operate the CSI at different output power operating points and to evaluate its loss for different configurations. The second objective is to use these operating points to verify (19) and, in the process, provide estimates of the three principal loss components of the inverter (excluding the dc-link inductor). Fig. 11 shows the total loss estimates for the three CSI configurations evaluated using (19) together with its estimated loss distribution. These figures also plot the error of the CSI loss prediction using (19) compared to the total loss measured experimentally. 17 out of 18 of these loss predictions result in errors less than 5%, with the highest errors at light load.

These loss component estimates make it possible to develop better estimates of the CSI efficiency improvement that can be expected using monolithic bidirectional switches, as discussed in the next section.

C. 3 kW Current Source Inverter Integrated Motor Drive

Fig. 12(a)–(g) show the fabricated 3 kW CSI-IMD and its forming components while Fig. 12(h) shows a simplified schematic of the complete system. The system comprises of a CSR and a CSI in a back-to-back configuration, as also identified in the power and sensor boards [see Fig. 12(c) and (d)]. The front-end CSR regulates the dc-link current magnitude and operates in a buck mode, allowing the IMD to operate at low-speeds where the back-EMF is lower than the grid input voltage. A low profile 3 W cooling fan is integrated into the motor endcap to provide cooling for the power electronics. The

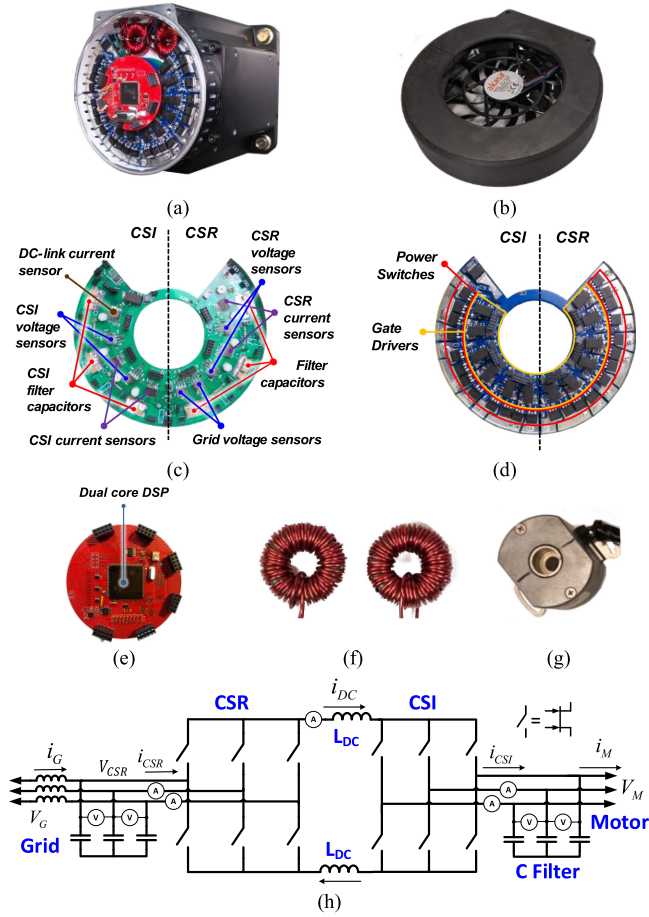


Fig. 12. Fully-assembled CSI-IMD and its components. (a) Fully assembled demonstrator with endcap removed. (b) Endcap with 3 W integrated cooling fan. (c) Sensor board. (d) Power board. (e) Controller board. (f) DC-link inductors. (g) Incremental encoder. (h) Simplified schematic of CSI-IMD.

controller TMS320F28379D is used to generate the gate signals and to implement the inverter control algorithm. The fabricated IMD intrinsically incorporates an under-damped LC circuit on both sides of the converter due to the combination of the filter capacitors with the grid inductance (on the rectifier side) and the machine inductance (on the motor side). Therefore, the filter capacitor values were carefully selected, and virtual resistances were included in the control law to ensure stability and improve the system's robustness, as discussed in [27] and [28].

The sensor board is composed of high-bandwidth Hall effect current sensors (ACS733KLATR-40AB-T) and galvanically-isolated voltage sensors (SI8920BC-IP). Only two line-to-line voltages and two phase currents are measured per converter (i.e., CSR and CSI), as identified in Fig. 12(h). The remaining line-to-line voltage and phase current are calculated based on Kirchhoff's laws. An additional current sensor is used for measuring the dc-link current. The power devices and gate driver configuration are the same as previously presented in Section V. Two 160 μ H dc-link *Edge* powder core (59083A2) inductors were designed following the methodology presented in [29] and each inductor has 55 turns and 37 m Ω of dc-resistance (@ 25 $^{\circ}$ C). The ten-pole SPM motor has a line-to-line resistance of 0.38 Ω ,

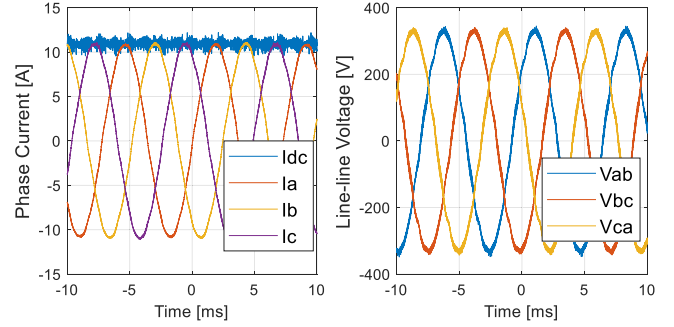


Fig. 13. Measured output voltage and output current of CSI-IMD implemented with FQ switches under three-step operation for 3 kW operation at 1652 RPM.

a line-to-line inductance of 5.9 mH, and a line-line back-EMF voltage of 137 $V_{rms}/kRPM$.

Fig. 13 shows the motor phase voltage and currents delivered by the FQ-switch-based CSI-IMD during 3 kW nominal operation at 1652 RPM (137.7 Hz). The motor currents have negligible ripple because the inverter PWM harmonic components are filtered by the second-order LC filter consisting of the machine inductance and the CSI's output filter capacitors. Therefore, the motor PWM loss is typically very low and almost insensitive to the PWM harmonics for switching frequencies that are above ~ 10 times the LC resonant frequency of approximately 2 kHz. The measured efficiency of the CSI driving the motor (excluding rectifier stage, inductor losses, auxiliary power supplies, and cooling fan) is 98.1% during 3 kW nominal operation at a switching frequency of 100 kHz when using the three-step commutation method. The measured efficiency when including the inductor and the cooling fan losses was 97.7%.

Due to the CSI and CSR stages' symmetry, the CSI loss analysis can also be applied to the CSR and the only major difference is that the grid will define the input voltage of the CSR that otherwise is defined by the machine back-EMF on the CSI side.

VI. PROJECTED EFFICIENCY OF FUTURE CSI-IMDS USING MONOLITHIC BIDIRECTIONAL SWITCHES

As previously mentioned, monolithic bidirectional switches can be particularly appealing when designed in a common-drain configuration; which substantially reduces their on-resistance and size [8], [9]. Unfortunately, these monolithic switches are still not commercially available, even though prototype switches have already been produced by multiple semiconductor manufacturers and research centers [8], [9].

Fig. 14(a) presents an efficiency map projection for a future CSI-IMD with a 3 kW power rating that use monolithic bidirectional switches operating using the three-step commutation method. The projection is made based on the empirical (19) derived from the experimental results obtained from the 3 kW fabricated CSI-IMD. However, to account for the lower resistance of the monolithic devices' channel resistance, the term R_{chan} has been set to 50% of the corresponding value for the hybrid bidirectional switches resistance that are implemented with two discrete devices. This reduction of $\sim 50\%$ of the total on-resistance and power device area made possible by using

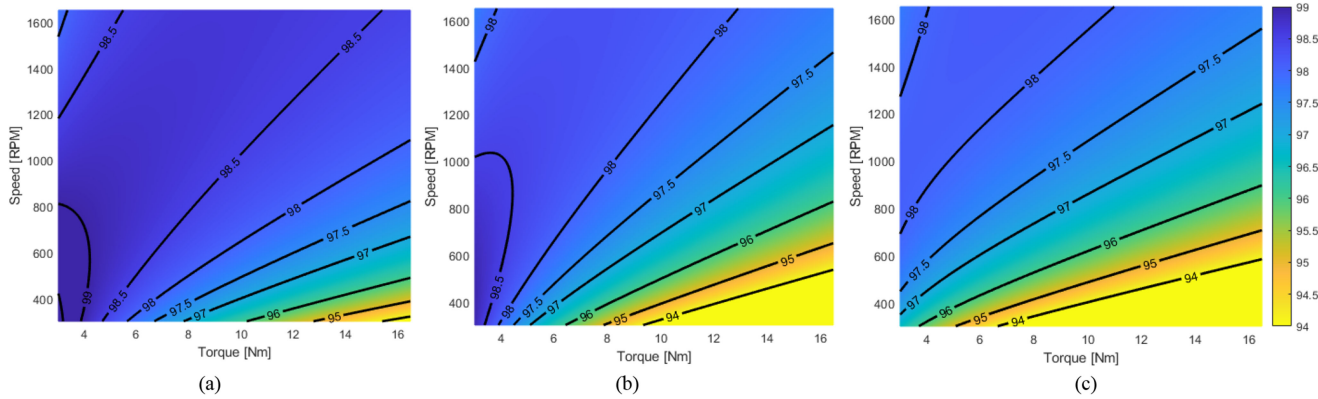


Fig. 14. Predicted CSI efficiency maps for 3 kW unit based on empirical formula (19). (a) Efficiency for CSI-IMD implemented with monolithic bidirectional switches with ON-state-resistance equal to half the ON-state resistance of hybrid bidirectional switches. (b) Efficiency of CSI-IMD using the hybrid FET + FET switch configuration using three-step operation. (c) Efficiency of CSI-IMD using conventional FET + diode switch configuration.

monolithic FQ switches is consistent with results reported in the literature [10].

In contrast to the preceding section, these efficiency maps include losses associated with the *Edge* powder core dc-link inductor. The inductor losses were estimated based on the methods provided in [29] and used the inductor core material curves provided by the manufacturer [30]. High-frequency ac loss components (100 kHz) including skin effect and hysteresis losses are also included in the loss estimates. The torque and speed values are based on the same demonstrator SPM motor described above.

For comparison, Fig. 14(b) and (c) show the efficiency map for the same CSI-IMD using the hybrid FET + FET switch configuration with three-step commutation and for a CSI using conventional FET + diode switches, respectively.

These plots show similar trends for the efficiency that drops considerably for low speeds and high torques. At low speeds, the output power is low at high torque values due to the high conduction losses produced by the high currents needed to produce the torque. Nevertheless, these plots show that adoption of monolithic FQ switches can substantially boost the CSI efficiency over a wide range of torque-speed operating points that is achieved by reducing the conduction loss, which is the main loss component in the CSI. By comparing Fig. 14(a) against Fig. 14(b) and (c), it can be seen that the efficiency improvement obtained when using monolithic FQ switches is more significant at low-speed and high torque operating conditions.

As noted in Section V, the FET + FET configuration has higher switching losses than the FET + diode configuration. Therefore, if the switching frequency is high enough, the traditional FET + diode configuration can become a more efficient option. In case monolithic bidirectional switches are used, this crossover frequency increases to 470 kHz for the baseline operating conditions listed in Table III compared to 220 kHz when using hybrid bidirectional switches implemented with discrete power devices. Such an increase in the switching frequency up to 470 kHz with reduced losses compared to traditional CSIs is highly desirable for improving the power electronics power density by shrinking passive components such as the dc-link inductor as well as the cooling system.

VII. CONCLUSION

This article has presented and tested four-step and three-step commutation methods that enable the use of FQ WBG switches in CSIs, allowing substantial reduction of conduction losses in the power devices at the cost of a small increase in the switching losses. Experimental results confirm that the use of FQ switches can significantly boost the efficiency of CSIs topology. The three-step commutation method minimizes the switching loss overhead of using FQ switches by reducing the required time interval when the FQ switches operate in their lossy switch-diode mode by 3:1. The method does not add any mass, volume, or cost penalties.

Efficiency maps of several CSI topologies have been presented, highlighting the advantage of the three-step CSI implemented with monolithic bidirectional switches that can achieve an efficiency higher than 98.5% for most of the torque-speed range, while, this efficiency is mostly below 97.5% for the baseline FET + diode topology.

The datasheets of the newest generation of SiC power devices indicate that their losses are significantly reduced compared to the devices used for this investigation. As a result of this rapid progress, there is a reason for confidence that the CSI efficiency values reported in this article for 3 kW operation will be raised by 0.5 to 1% when these new devices are introduced. Ongoing research is aggressively pursuing these improvements, which will be reported in future papers.

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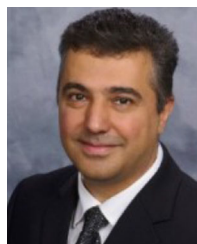
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